

## DATA-260 HOMEWORK 5 – s0036

Repository: data260-0036

SID4: 0036      PORT\_BASE: 8036      PREFIX: s0036      SEED: 36

VERIFY\_SEED: 260036      DOMAIN\_ID: 4

Hardware: MacBook Pro, Apple M5 Pro, 24 GB

Local model: qwen3:4b through Ollama

### PART 1 – API AND REDUX

VulnerabilityReport is the primary entity and Advisory is the related entity. The SQLAlchemy schema includes identifiers, unique advisory codes, timestamps, a report advisory foreign key, available\_count defaulting to one, and restricted related-record deletion. FastAPI exposes paginated advisory CRUD, report CRUD, and reports-for-advisory. The React data layer uses Redux Toolkit thunks for fetch/create/update/delete and a Redux slice for state updates.

### PART 2 – MCP SERVERS

meals\_server.py provides search\_meals\_by\_name, meals\_by\_ingredient, random\_meal, and meal\_details over TheMealDB with timeout/retry handling. domain\_mcp\_server.py provides search, detail, and aggregate with the shared {ok, data, error} envelope. Both servers use STDIO and Python logging for diagnostics.

### PART 3 – CONTRACTS AND FAULTS

The domain tools validate invalid query, id, limit, and severity inputs. Fault injection uses VERIFY\_SEED 260036, three attempts, a five-second timeout, and bounded exponential backoff. Raw records for 0%, 20%, and 50% rates are in reports/hw05/raw/fault\_injection.csv and .json; the generated summary is fault\_summary.json.

### PART 4 – SAFE ENTRY POINT

execute\_tool is the only agent-facing entry point and always returns a JSON string with the shared envelope. The offline runner has eight assert-based tests, including valid and invalid calls for all three tools.

### PART 5 – SAFETY AND AGENT

The safety rule blocks aggregate requests for restricted unavailable inventory. run\_agent enforces max\_steps and writes each tool call and final stop reason to raw/agent\_runs.jsonl. Four scenario summaries are in raw/agent\_scenarios.json. Reflection and AI-use disclosure are included in REFLECTION.md and AI\_USE.md.

### VERIFICATION

Offline tests: 8/8 passed. Frontend production build: passed. Run PYTHONPATH=. python3 scripts/verify\_hw5.py to regenerate verification.json.